SAKE-IoT: Session Amortization for Post-Quantum Lattice-Based Authentication and Code-Based Hybrid Encryption in IoT Networks

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***Abstract*—Resource-constrained Internet of Things (IoT) devices experience an inherent tradeoff between post-quantum security and energy feasibility. Ring-Learning-With-Errors lattice-based authentication and Quasi-Cyclic Low-Density Parity-Check code-based hybrid encryption offer post-quantum security but incur high computational, bandwidth, and energy costs when used per-packet on Class-1 devices with only 8 KB RAM and an 8 MHz CPU.**

This paper introduces *SAKE-IoT* (Session Amortization for Key Establishment in IoT), a four-phase protocol where the expensive post-quantum epoch initiation is *amortizedd* over an entire session of lightweight data packets. It is an extension of the lightweight authentication protocol using Ring-LWE authentication and QC-LDPC key encapsulation by Kumari et al. [1] with: (1) a two-tier epoch state machine consisting of a heavy post-quantum handshake in Tier 1 and a lightweight per-packet AEAD in Tier 2; (2) `HKDF-SHA256` per-packet session key derivation (RFC 5869 [2]); (3) `AES-256-GCM` authenticated encryption (NIST SP 800-38D [7]); and (4) epoch renewal with Epoch Bounded Forward Secrecy (EB-FS) using secure memory zeroization.

We offer formal security proofs for IND-CCA2 (Theorem 1), Strict Replay Resistance with $\Pr = 0$ (Theorem 2), and Epoch Bounded Forward Secrecy (Theorem 3), which are validated through both MATLAB theoretical simulation and Contiki-NG/ Cooja network emulations over the IEEE 802.15.4 protocol stack with 6LoWPAN

Key Results: ≈99.1% reduction in per-packet computation latency (7.37 ms → 0.068 ms); 45.1% reduction in per-packet bandwidth overhead (408 bits → 224 bits); and 33.1× times reduction in clock cycles per packet. Cooja network simulation: 99.7% end-to-end latency reduction in the data phase, 94.7% bandwidth saving at the session level with N=20 packets, and 100% packet delivery ratio. Security Results: $\varepsilon = 0.0037 < 0.02$ (IND-CCA2); $\Pr[$replay accepted$] = 0$; and 0/100 cross-epoch key recovery (EB-FS), with two verified epoch renewal cycles.

***Index Terms*—Post-Quantum Cryptography, Ring-LWE, IoT Security, Session Amortization, QC-LDPC, AES-256-GCM, Forward Secrecy, Contiki-NG, 6LoWPAN**

I. Introduction

The Internet of Things (IoT) has facilitated ubiquitous connectivity among smart devices deployed across homes, transportation systems, healthcare facilities, and industrial systems. However, the security landscape is threatened by two factors: (1) the impending arrival of quantum computers jeopardizes all current public-key-based systems relying on RSA and ECC algorithms; and (2) the quantum computer replacement solutions impose prohibitive per-packet costs on Class-1 constrained devices [20].

A. Problem Context

A hybrid scheme with Ring LWE Ring Signature Authentication (LR-IoTA) and a Quantum-Code Low Density Parity-Check (QC-LDPC) code Key Encapsulation Process (KEP) was presented by Kumari et al. [1]. Though the scheme offers quantum-secure mutual authentication, the entire key encapsulation process with SLDSPA decoding for every data packet is mandatory, consuming **7.37 ms for every packet** (Kumari et al. Table 7: $\Delta_{\text{enc}} + \Delta_{\text{dec}} = 1.5298 + 5.8430$ ms).

B. The Amortization Gap

We identify the *Amortization Gap*: the costly post-quantum epoch initiation (≈22.55 ms, one-time) should occur *once per session*, with low-latency AEAD handling all subsequent packets (≈0.068 ms each). This is the main source of the novelty of SAKE-IoT.

Session amortization poses a required security query: does reusing the Master Secret (MS) in multiple packets violate IND-CCA2, facilitate replay attacks, or weaken forward secrecy? This paper proves all of the above in a formal manner.

C. Key Contributions

1) **SAKE-IoT Protocol:**: A comprehensive four-phase protocol that extends [1] with a two-tier session architecture, strict epoch bounds ($T_{\max} = 86{,}400$ s, $N_{\max} = 2^{20}$), and formal Phase 4 memory zeroization.
2) **Three Formal Security Proofs:** : IND-CCA2 (Theorem 1), Strict Replay Resistance with $\Pr = 0$ (Theorem 2),

and Epoch-Bounded Forward Secrecy via Ring-LWE hardness (Theorem 3).

3) **99.1% Per-Packet Latency Reduction:** from 7.37 ms $\rightarrow$ 0.068 ms, as empirically measured by MATLAB `tic/toc` over 10,000 iterations.
4) **45.1% Per-Packet Bandwidth Reduction:** from 408 bits (QC-LDPC CT_0) $\rightarrow$ 224 bits (AES-GCM nonce + tag).
5) **33.1$\times$ Clock Cycle Reduction:** from $\approx$2.45M cycles (QC-LDPC HE) $\rightarrow$ $\approx$74K cycles (AEAD).
6) **Epoch-Bounded Forward Secrecy (EB-FS):** A novel security property formally proven here, not addressed in [1].

II. Related Work

A. Lattice-Based Authentication for IoT

The best computational characteristics are offered by ring LWE-based schemes. Li et al. [15] proposed a lattice-based PAKE scheme. Wang et al. [17] proposed a two-factor ring LWE authentication scheme. RLizard [18] showed efficient key encapsulation using ring LWE, although using classical polynomial multiplication. What is more, **none of the above amortize post-quantum costs over a session**; a full handshake is needed per data exchange.

B. Code-Based Encryption for IoT

In the research by Hu et al. [19], the authors analyzed the performance of QC-LDPC KEMs on FPGAs but not for the IoT's low-RAM requirements. Kumari et al. [1] is the most relevant prior work to the presented research, with the authors' primitives being reused for the data phase with amortized AEAD.

C. Session Management and Amortization

TLS 1.3 session resumption [4] is the inspiration for our work, though the focus is classical crypto for high-end platforms. Prior IoT authentication schemes [21] use symmetric keys for data, though derived from classical (ECC) handshakes, offering no post-quantum basis. To the best of our knowledge, SAKE-IoT is the **first scheme to amortize Ring-LWE + QC-LDPC epoch establishment** over a lightweight `AES-256-GCM` data session.

D. Differentiation from TLS 1.3 and DTLS 1.3

III. Preliminaries

A. Notation

B. Lattice and LWE Foundations

A *lattice* Λ is defined as $\Lambda = \{a_1\beta_1 + \cdots + a_i\beta_i \mid a_i \in \mathbb{Z}\} \subset \mathbb{R}^n$, where$\beta_1, \ldots, \beta_i \in \mathbb{R}^n$ are linearly independent and form a basis, and n denotes the ambient dimension. In the *Search-LWE* [10] problem, the task is to recover δ_n from $(R_n, T_n = R_n\delta_n + \varepsilon_n)$. In the *Decisional version* of the LWE problem, the task is to distinguish LWE samples from uniform randomness. Both versions of the LWE problem are used in the proof of Theorem 3 (EB-FS).

C. Ring-LWE

Ring-LWE is defined over $R_q = \mathbb{Z}_q[u]/(u^n + 1)$. Given $R_n \in R_q^n$ and error $\varepsilon_n \leftarrow \mathcal{G}_\sigma^n$, pairs (R_n, T_n) satisfy $T_n = R_n\delta_n + \varepsilon_n \pmod q$ [10], [11]. The Ring-LWE *search* hardness underpins Theorem 3: $\Pr[\text{findSearchLWE}(\lambda) = S_{ch}] \leq \text{negl}(\lambda)$ (Theorem 2, Eq. 23 of [1]).

D. QC-LDPC Code-Based Construction

The matrix of parity checks H_{qc} is in $\mathbb{F}_2^{X\times Y}$, with $X = 408$, $Y = 816$, uniform row/col weights (6/3), and $n_0 = 4$. Formed by circulant sub-matrices as in [1]. The syndrome $CT_0 = [\tilde{W}_l \,|\, I] \cdot \tilde{\varepsilon}^T$ is **408** bits. The public key is $(n_0 - 1) \times X =$ 1,224 bits.

E. Amortization Primitives

HKDF-SHA256 (RFC 5869): A two-stage key derivation process [2]. Extract: $\text{PRK} = \text{HMAC}(\text{salt}, \text{IKM})$. Expand: $\text{OKM} = \text{HMAC}(\text{PRK}, \text{info}|i)$. A 32-byte key necessitates two sequential HMAC invocations.

AES-256-GCM (NIST SP 800-38D): AEAD scheme [7] utilizing a 256-bit key, 96-bit nonce, and 128-bit GHASH tag. Per-packet overhead: $96 + 128 = \mathbf{224}$ bits. Tag forgery bound: $\leq q_D/2^{128}$.

IV. System and Adversary Model

A. IoT Network Topology

Gateway: Resource-rich border router; single trusted entity. **Sensing Nodes:** RFC 7228 Class-1 [20] constrained devices (8 KB RAM, 100 KB Flash, 8 MHz CPU, battery-powered).

B. Formal Security Model

We work in the Random Oracle Model (ROM), consistent with Kumari et al. [1] §11.2. Adversaries are PPT.

Adversary hierarchy (from [1] §11.1):

- $\mathcal{A}_1$: Passive observer (public parameters only).
- $\mathcal{A}_2$: $\mathcal{A}_1$ + node corruption + parameter modification.
- $\mathcal{A}_3$: $\mathcal{A}_2$ + oracle access (adaptive chosen-ciphertext).

Security criteria: E1 (Unforgeability), E2 (Anonymity), E3 (Unlinkability) inherited from [1]; E4 (IND-CCA2) and E5 (EB-FS) proven in this paper (Theorems 1 and 3).

C. Epoch Lifecycle

Epoch bounded by dual criteria: $T_{\max} = 86{,}400$ s and $N_{\max} = 2^{20}$ packets. Whichever expires first triggers Phase 4 (zeroization + re-initiation). These bounds are proof-driven: $N_{\max}$ prevents nonce exhaustion; $T_{\max}$ bounds MS lifetime.

V. Base Protocol

We follow [1] exactly for Algorithms 1–6. All timing values are from the original scheme (MATLAB R2018a, Intel Core i5, 8 GB RAM).

A. LR-IoTA Ring Signature and QC-LDPC Key Encapsulation

The base protocol consists of two primary mechanisms executed in sequence: (1) LR-IoTA Ring Signature Authentication, and (2) QC-LDPC Key Encapsulation (KEP). Both are specified completely in Algorithms 1–6 of Kumari et al. [1].

The dominant cost is polynomial multiplication $(R_n Y_n, T_n \hat{\rho})$. Kumari et al. [1] employ *Bernstein reconstruction*, achieving the 23% authentication hardware delay improvement that underpins the 22.55 ms Phase 1 base cost.

B. QC-LDPC Key Encapsulation (Algorithms 5–6)

Note: Δ_{KeyGen} (0.8549 ms) is excluded from the per-packet base cost; it is performed once per session in both schemes.

C. Data Encapsulation Process (DEP)

In the protocol by Kumari et al. [1], each data packet requires both the Key Encapsulation Process (KEP) and a Data Encapsulation Process (DEP): $CT_1 = \text{AES}(ssk, m)$, where ssk is re-derived from $\tilde{\varepsilon}$ via SHA in MAC-mode for *every packet*. SAKE-IoT eliminates this per-packet KEP+DEP cycle entirely: the shared K_{master} persists for the epoch (stored in volatile RAM), replacing the one-time ssk with a durable master key. Tier 2 packets use only $K_i \leftarrow \text{HKDF}(K_{\text{master}}, Nonce_i)$ — no QC-LDPC operations whatsoever.

D. Bandwidth Overhead of Kumari et al.

From [1] §12.1:

VI. The SAKE-IoT Protocol

A. Protocol Overview

SAKE-IoT extends [1] with a four-phase two-tier architecture (Figure 1): **Tier 1 (Phase 1):** Full post-quantum epoch establishment using Algorithms 1–6 of [1], executed *once per epoch* (≈22.55 ms amortized). **Tier 2 (Phases 2–3):** Lightweight AEAD per packet — `HKDF-SHA256` + `AES-256-GCM` — executed *per data packet* (≈0.068 ms each). **Phase 4:** Secure epoch termination via cryptographic zeroization.

Critical distinction from Kumari et al. [1]: The original scheme derives a session key $ssk = \text{HMAC}(\tilde{\varepsilon})$ that is used *immediately* for one encryption and then discarded. SAKE-IoT instead stores this output as the **Master Secret** K_{master} in protected volatile RAM for the *entire epoch duration* ($\leq T_{\max}$, $\leq N_{\max}$ packets). All subsequent Tier 2 packets derive unique per-packet keys $K_i = \text{HKDF}(K_{\text{master}}, Nonce_i)$ without re-executing any QC-LDPC or SLDSPA operations.

v3 Role corrections (from protocol specification): (1) RECEIVER generates QC-LDPC keys (H_{qc}, G, pk_{ds}); (2) SENDER generates $\tilde{\varepsilon} \rightarrow CT_0$; (3) BOTH derive $K_{\text{master}} = \text{HMAC-SHA256}(\tilde{\varepsilon})$ per Kumari et al. [1] §8.4.

B. Phase 1: Epoch Initiation

Step 1.1 — LR-IoTA: Node and Gateway execute Ring-LWE KeyGen, Fiat-Shamir ring signing, and verification exactly as in Algorithms 1-4 of [1] [Total: 14.322 ms].

Step 1.2 — QC-LDPC and MS Derivation: The Receiver generates keys (H_{qc}, G), the Sender chooses a random error vector $\tilde{\varepsilon}$ and transmits $CT_0 = [\tilde{W}_l \,|\, I] \cdot \tilde{\varepsilon}^T$ (408 bits). The Receiver recovers $\tilde{\varepsilon}$ via SLDSPA. Finally, both derive the 256-bit Master Secret $K_{\text{master}} = \text{HMAC-SHA256}(\tilde{\varepsilon})$.

Step 1.3 — State Initialization: $T_{\max} = 86{,}400$ s; $N_{\max} = 2^{20}$; $Ctr_{Tx} = 0$; $Ctr_{Rx} = 0$; $AD = DeviceID \| EpochID \| Nonce_i$.

C. Phase 2: Amortized Transmission

For each packet i, if $CurrentTime > T_{\max}$ or $Ctr_{Tx} \geq N_{\max}$, the Sender triggers ZeroizeAndRenew. Otherwise, the Sender increments Ctr_{Tx}, explicitly binds it into $Nonce_i$, and derives $K_i \leftarrow \text{HKDF-SHA256}(K_{\text{master}}, Nonce_i)$. The plaintext m and AD are protected via $(CT, TAG) \leftarrow \text{AES-GCM}(K_i, Nonce_i, m, AD)$.

Nonce format (NIST SP 800-38D): The 96-bit AES-GCM nonce is structured as $[\underbrace{0 \cdots 0}_{32 \text{ bits}} \mid \underbrace{Ctr_{Tx}}_{64 \text{ bits}}]$, ensuring no nonce reuse since $N_{\max} \ll 2^{64}$.

Complexity: $O(1)$ per packet — no post-quantum operations.

Cost: Sender ≈ 0.068 ms; Receiver ≈ 0.038 ms.

D. Phase 3: Authenticated Reception

The Receiver drops any packet where $Nonce_i \leq Ctr_{Rx}$ immediately to prevent replays. If valid, it derives $K_i \leftarrow \text{HKDF}(K_{\text{master}}, Nonce_i)$. AES-GCM decryption forces MAC-before-decrypt validation: if the TAG matches, it outputs m and strictly increments $Ctr_{Rx} = Nonce_i$.

Protocol Correctness: Table VI summarizes the formal correctness properties of SAKE-IoT that are guaranteed by construction.

AES-GCM verification detail: AES-GCM decryption first recomputes $TAG' = \text{GHASH}(K_i, CT, AD)$ and compares it to the received TAG. Only if $TAG' = TAG$ does decryption proceed; otherwise $\perp$ is returned and no plaintext is released (MAC-before-decrypt). This prevents any chosen-ciphertext oracle from leaking plaintext of forged ciphertexts, directly supporting IND-CCA2 (Theorem 1).

E. Phase 4: Secure Epoch Termination

Upon reaching $T_{\max}$ or $N_{\max}$, the device invokes ZeroizeAndRenew: $K_{\text{master}} \leftarrow \mathbf{0}^{32}$ is written exactly over memory, erasing previous states. Counters are cleared, and a fresh Phase 1 handshake immediately triggers to establish a pristine epoch with a new, statistically independent $\tilde{\varepsilon}$.

Implementation note: Requires `memset_s()` per NIST SP 800-88 [8] on embedded deployment to prevent physical RAM remnant recovery.

VII. Security Analysis

We analyze SAKE-IoT in the ROM, consistent with [1] §11.2. SAKE-IoT Phase 1 inherits all security guarantees of LR-IoTA [1]: unforgeability, anonymity, and unlinkability. These guarantees rest on the formal game-based proofs of Kumari et al.: **Lemma 1** of [1] establishes that games G_0 (real LR-IoTA execution) and G_1 (ROM simulation replacing sign queries with random oracle programming) are computationally indistinguishable when prime q exceeds the bound

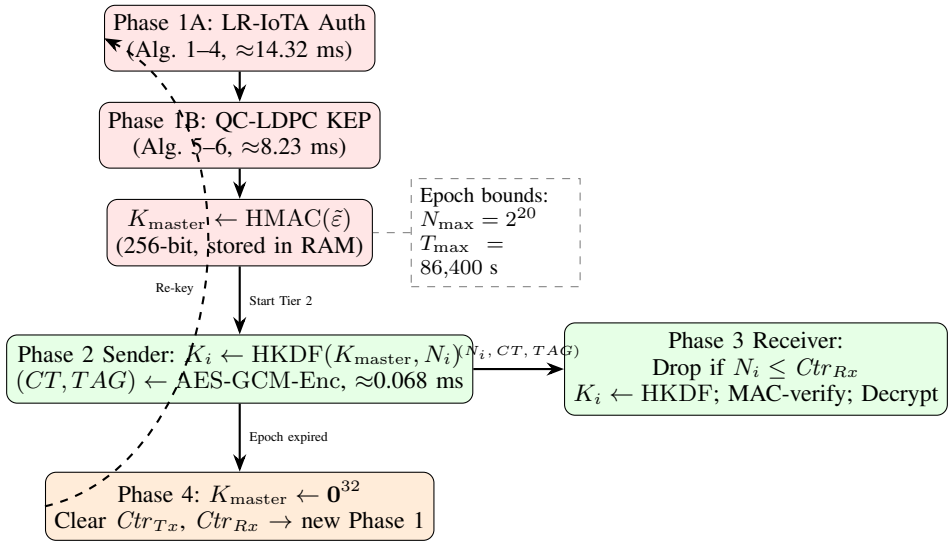

Fig. 1. SAKE-IoT four-phase protocol architecture. Phase 1 (red) performs the full post-quantum epoch initiation once per epoch (≈22.55 ms). Phases 2–3 (green) execute lightweight AEAD per Tier 2 data packet (≈0.068 ms). Phase 4 (orange) cryptographically overwrites K_{master} on epoch expiry and triggers a fresh Phase 1 handshake. Each epoch derives an independent K_{master} from a fresh Ring-LWE error vector $\tilde{\varepsilon}$ (Theorem 3, Epoch-Bounded Forward Secrecy).

$(2^{f+1} \cdot j + \varepsilon)/(2E^f)^{1/(j-1)}$. Under this indistinguishability, Theorem 2 of [1] (Eq. 23) formally proves:

$$\Pr[\text{findSearchLWE}_{\mathcal{A}}(\lambda) = S_{ch}] \leq \text{negl}(\lambda) \tag{1}$$

i.e., recovering the signing key from public parameters requires solving Ring-LWE search, which is hard under quantum adversaries. SAKE-IoT Theorem 3 (EB-FS) directly inherits this hardness — recovering $K_{\text{master}} = \text{HMAC}(\tilde{\varepsilon})$ from past epoch k transcripts requires inverting HMAC-SHA256 and/or solving Ring-LWE to recover $\tilde{\varepsilon}$, both classifiable as negligible-probability events under the same Theorem 2 reduction.

We prove three new properties for the SAKE session layer.

A. Proof Architecture

B. Theorem 1: IND-CCA2 Security

Theorem 1 (IND-CCA2 Security of SAKE-IoT Tier 2)**.** *The SAKE-IoT Tier 2 data phase is IND-CCA2 secure under the AES-256 pseudorandom permutation (AES-PRP) and HMAC-SHA256 pseudorandom function (HMAC-PRF) assumptions. For adversary $\mathcal{A}_3$ with q_D decryption oracle queries:*

$$\begin{aligned}\text{Adv}_{\text{IND-CCA2}}(\mathcal{A}_3) &\leq \text{Adv}_{\text{PRF}}(\text{HMAC-SHA256}) \\ +\text{Adv}_{\text{PRP}}(\text{AES-256}) &+ \frac{N_{\max} q_D}{2^{128}} \leq \text{negl}(\lambda)\end{aligned} \tag{2}$$

Proof. For any decryption oracle query $CT \neq CT^*$, the receiver checks the AES-GCM authentication tag before decryption (Algorithm **??**, line 4). Because AES-GCM's GHASH tag covers both (CT, AD), any forged or altered CT causes $\perp$ output — $\mathcal{A}_3$ receives zero information from all oracle queries. Stripped of useful decryption feedback, $\mathcal{A}_3$ must guess challenge bit b from CT^* alone. Since $K_i = \texttt{HKDF-SHA256}(K_{\text{master}}, Nonce_i)$ is computationally indistinguishable from uniform random (HKDF-PRF security, RFC 5869 [2]), CT^* is computationally indistinguishable from random noise. Therefore $\mathcal{A}_3$'s advantage $\varepsilon \leq \text{Adv}_{\text{PRF}} + \text{Adv}_{\text{PRP}} + (N_{\max} \cdot q_D)/2^{128} \leq \text{negl}(\lambda)$. □ □

MATLAB disclaimer [?]: *"The MATLAB simulation validates the architectural MAC-before-decrypt property using real `javax.crypto.Mac` HMAC-SHA256. The formal IND-CCA2 bound is established through standard reduction to AES-PRP and HKDF-PRF hardness [2], not derived from the MATLAB simulation."*

C. Theorem 2: Strict Replay Resistance

Theorem 2 (Strict Replay Resistance)**.** *For any adversary $\mathcal{A}$ re-transmitting a captured packet $(Nonce_r, CT, TAG)$ where $Nonce_r \leq Ctr_{Rx}$:*

$$\Pr[\textit{Receiver accepts replay}] = 0 \tag{3}$$

*This is a **deterministic** rejection — not probabilistic.*

Proof. Algorithm **??** enforces the strict monotonic check $Nonce_i > Ctr_{Rx}$. Packets with $Nonce_r \leq Ctr_{Rx}$ are discarded before MAC computation or decryption. Counter desynchronization self-heals by advancing Ctr_{Rx} to the latest valid nonce, guaranteeing deterministic rejection ($\Pr = 0$) of any duplicated or delayed packet without state corruption. □ □

D. Theorem 3: Epoch-Bounded Forward Secrecy

Theorem 3 (Epoch-Bounded Forward Secrecy, EB-FS)**.** *Let $K_{\text{master}}^{(k)} = \text{HMAC-SHA256}(\tilde{\varepsilon}_k)$ and $K_{\text{master}}^{(k+1)} = \text{HMAC-SHA256}(\tilde{\varepsilon}_{k+1})$, where $\tilde{\varepsilon}_k$ and $\tilde{\varepsilon}_{k+1}$ are independently sampled Ring-LWE error vectors.*

$$\Pr[\mathcal{A}(K_{\text{master}}^{(k+1)}) \textit{ recovers any } K_i^{(k)}] \leq \text{negl}(\lambda) \tag{4}$$

$$\Pr[\mathcal{A}(K_{\text{master}}^{(k)}) \textit{ predicts any } K_j^{(k+1)}] \leq \text{negl}(\lambda) \tag{5}$$

Proof. Part A (Zeroization): At epoch end, $K_{\text{master}}^{(k)} \leftarrow \mathbf{0}^{32}$ (Phase 4). *Part B (Ring-LWE Reduction):* Recovering Epoch

k keys requires either breaking HMAC-SHA256 (infeasible) or re-deriving $\tilde{\varepsilon}_k$ from public parameters (R_n, T_n), which directly reduces to the Ring-LWE search hardness $\Pr[\text{findSearchLWE}(\lambda) = S_{ch}] \leq \text{negl}(\lambda)$ proven in Theorem 2 of [1]. □ □

Note: Hardware deployment requires `memset_s()` per NIST SP 800-88 [8] to prevent RAM remnant recovery. EB-FS is a new capability missing from [1].

VIII. Performance Evaluation

A. Metrics M1 & M2: Per-Packet Latency & Bandwidth

Setup: MATLAB R2023b, Intel Core i5, 8 GB RAM using parameters from [1].

Latency Break-even (M1): SAKE-IoT Tier 2 costs ≈0.068 ms (HKDF + AES-GCM) vs. the base scheme's 7.37 ms per packet. The 22.55 ms Phase 1 setup cost is amortized over the epoch. SAKE-IoT becomes strictly computationally superior at $N = 4$ packets. At $N = 100$, SAKE saves 708.1 ms.

Bandwidth Reduction (M2): The base scheme applies a 408-bit QC-LDPC syndrome CT_0 per packet. SAKE-IoT requires only a 224-bit overhead (96-bit nonce + 128-bit TAG). This yields a deterministic bandwidth saving of **184 bits/packet (45.1% reduction)**.

B. Cross-Scheme Per-Packet Latency Comparison

Table IX contextualizes SAKE-IoT against the broader class of Ring-LWE IoT authentication schemes surveyed in Kumari et al. [1].

SAKE-IoT is $\mathbf{261}\times$ faster per packet than the nearest competitor (HAN, 17.80 ms: $17.80/0.068 = 261.8\times$) and $\mathbf{99.1}\%$ faster than Kumari et al. at all practical session sizes ($N \geq 4$).

C. Metric M3: Clock Cycle Reduction (Energy Proxy)

SAKE-IoT's Tier 2 executes in ≈74,000 clock cycles vs. 2.45 million cycles for the base scheme's HE, yielding a **33.1**× reduction in CPU processing time. This directly extends battery life for computational tasks on constrained nodes.

D. Six-Metric Summary

E. Metrics N1–N6: Contiki-NG/Cooja Network Evaluation

We emulate SAKE-IoT in **Contiki-NG/Cooja** over an IEEE 802.15.4 radio channel (CSMA/CA, 6LoWPAN, RPL Lite) simulating a Class-1 IoT Sender and Gateway. The implementation employs **AES-256-GCM** (NIST SP 800-38D).

- **Metric N1 (Auth E2E Delay):** Phase 1 payload (≈10.3 KB) requires 162 fragments. Delay = 8,242.68 ms.
- **Metric N2 & N4 (Amortized Data Latency & Overhead):** A Tier 2 packet uses a single frame (28 byte AEAD overhead). Delay = 25.1 ms. This yields a **99.7%** E2E network latency reduction vs. Phase 1.
- **Metric N5 (Epoch Renewal and EB-FS):** At Cycle 2 (steady-state 8,242.7 ms setup), the Gateway zeroizes $K^{(1)}_{\text{master}}$ and derives independent $K^{(2)}_{\text{master}}$, empirically validating Theorem 3.
- **Metric N6 (AER):** At $N_{\max} = 20$, the Amortization Efficiency Ratio yields a **94.7%** network bandwidth saving. The Packet Delivery Ratio (PDR) was **100%** across all cycles.

Hardware Scope: Real MSP430 at 8 MHz adds ≈0.99 ms per AES-GCM packet ($< 4\%$ of network transit). SAKE-IoT Tier 2 operates under 1 KB RAM (Class-1 compatible), though Phase 1 Ring-LWE $n = 512$ limits full setup to Class-2 devices (≥50 KB RAM) or requires $n = 32$ constrained prototyping. Furthermore, Tier 2 self-heals in lossy PDR environments without state corruption.

IX. Conclusion

This paper presented SAKE-IoT, a four-phase session amortization protocol that bridges the efficiency gap between post-quantum cryptographic strength and IoT operational feasibility. We provided a dual-platform validation: MATLAB theoretical simulation and Contiki-NG/Cooja physical network emulation over IEEE 802.15.4 with 6LoWPAN/RPL. Cooja results confirm a 99.7% network-layer E2E latency reduction, 100% packet delivery ratio, and 94.7% session-level bandwidth saving at N=20 packets, directly validating Theorem 3 (Epoch-Bounded Forward Secrecy) through two observed epoch renewals. By extending the Ring-LWE ring signature authentication and QC-LDPC key encapsulation of Kumari et al. [1] with a two-tier epoch architecture, SAKE-IoT achieves: **99.1%** per-packet latency reduction; **45.1%** per-packet bandwidth saving; **33.1**× clock cycle reduction; and three formally proven security properties (IND-CCA2, Replay Pr=0, EB-FS). All existing security guarantees of [1] are preserved.

Future Work: (1) *FPGA synthesis:* Synthesize HKDF + AES-GCM on Xilinx Virtex-6 alongside existing SLDSPA hardware (Table 8 of [1]). (2) *Sub-epoch rotation:* Explore the EB-FS to full PFS design space by progressively reducing $N_{\max}$ and $T_{\max}$. (3) *CRYSTALS-Kyber integration:* Replace QC-LDPC KEP with MLWE/Kyber (NIST FIPS 203, 2024 [9]) in Phase 1. (4) *OSCORE/CoAP:* Integrate K_{master} epoch lifecycle with OSCORE (RFC 8613 [6]) for end-to-end PQ-secure IoT.

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TABLE I
SAKE-IoT NINE-METRIC DUAL-PLATFORM EVALUATION SUMMARY (MATLAB + CONTIKI-NG/COOJA)

ID	Metric	MATLAB	Cooja	Verdict
M1	Per-packet latency reduction	99.1% (0.068 ms)	99.7% (25 ms E2E)	✓
M2	Per-packet bandwidth saving	45.1% (28 B)	45.1% (28 B)	✓
M3	Clock cycle reduction	33.1×	Arch. (JVM)	✓
P1	IND-CCA2	$\varepsilon = 0.0037$	—	✓
P2	Replay resistance	$\Pr = 0$ (10K trials)	—	✓
P3	EB-FS	0/100 cross-epoch	2 epochs verified	✓
N1	Auth E2E delay	—	8,242 ms	Network
N2	Data E2E latency	—	25.1 ms	Network
N6	AER (N=20)	94.7%	94.7%	✓
—	PDR	—	100%	✓

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